Dehradun Vision 2025: Smart City Traffic Optimization Using YOLOv11, U-Net, and Deep SORT

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***Abstract*— This research presents an AI-powered traffic management system developed to solve Dehradun's urban mobility issues in alignment with Dehradun Smart City Vision 2025. It implements YOLOv11 for vehicle detection, U-Net for lane segmentation, Deep SORT for multi object tracking, and IoT-based adaptive signal control. Unlike fixed-timer systems, it adjusts signals in real-time considering traffic density, pedestrian movement, and other environmental factors. Using 3000 images from eight vehicle classes, the system outperformed YOLOv5, YOLOv7, and YOLOv8 with a 96.4% detection accuracy at 125 FPS, which also decreased waiting time by 38%. Detection accuracy nighttime for pedestrians boosted safety in low-visibility conditions to 92.5%. The system also reduced CO_2 emissions by 35% while maintaining 15ms latency with 5G transmission, achieving real-time decision making, reinforcing sustainability goals. The system outperformed other models in high-density traffic regions, and served as a proof of concept to show deep learning, IoT, and real-time optimization can improve urban mobility, congestion, and smart sustainable city planning.**

Keywords*— *Smart Traffic Management, YOLOv11, U-Net, Deep SORT, Adaptive Signal Control, IoT-Based Traffic Monitoring, Intelligent Transportation Systems (ITS), Multi Object Tracking (MOT).

I. INTRODUCTION

The capital city of Uttarakhand, Dehradun, along with other Indian cities, suffers from heavy traffic congestion due to the country's accelerating urbanization. Furthermore, congestion increases the rate of pollution as well as travel time. Fixed time controls and manual traffic supervision do not solve these problems cooperatively and have their limitations [1]. For urban mobility to have lower emissions alongside smoother traffic flow in Dehradun with the SMART CITY 2025 goals, an AI based modern traffic system needs to be implemented. Modern algorithms and deep learning neural networks, as well as computer vision technology provide excellent resources for monitoring traffic in real time, such as YOLOv11, a powerful object detection algorithm that is unmatched in speed and precision for discerning automobiles and pedestrians [2]. For the purpose of lane detection and examining traffic patterns within a single frame, U-Net, a global leader in image segmentation, proves to be invaluable. Moreover, the multi-purpose tracker Deep SORT increases safety and decreases obstruction in traffic by tracking multiple objects efficiently [3][4]. Traffic control can be greatly enhanced when AI is merged with IoT solutions, such as intelligent sensors, which alongside real time data manipulation expediently diminish holdups, emissions, and waiting period [5][6]. Cities like Pune and Bengaluru have already experienced notable changes in traffic congestion and environmental improvement through similar systems.

This research intends to design an intelligent traffic management system for Dehradun powered by AI with features including real-time traffic surveillance, automated signal control, and enforcement of the pedestrian right of way. Advanced algorithms such as vehicle detection using YOLOv11, lane segmentation with U-Net, and object tracking using Deep SORT will be utilized to maintain proper traffic movement and compliance with rules. Moreover, smart sensors will be able to anticipate traffic bottlenecks, support responsive changes to traffic control, and reduce waiting time IoT. Minimizing vehicle stopped time and emissions, enhanced environmental protection, and smarter city goal-2025-are some important objectives of this system. The research integrates DRL for adaptive signal control in Dehradun, performing real-time traffic analysis by merging YOLOv11, U-Net, and Deep SORT. Their findings indicate enhanced accuracy metrics and decreased wait times while maintaining Smart City Vision 2025 environmental objectives.

II. LITERATURE REVIEW

The addition of AI and DL technologies to Intelligent Systems for Transportation (IST) has enabled features such as real-time traffic monitoring, adaptive traffic signal control, traffic forecasts, and automation of violation processing, all of which have enhanced the mobility of modern cities. With the increasing volume of traffic, a combination of fixed-time signal control and manual observation has become obsolete, highlighting the need for AI solutions [7],[8]. Conversely, the integration of deep learning with IoT, edge devices, and 5G networks allow for the rapid processing of traffic data for timely response in the event of traffic congestion, urgent medical emergencies, or special consideration for emergency response vehicles [9]. Studies indicate that the implementation of intelligent systems for traffic management can reduce congestion by more than 30%, enhance road safety, and minimize vehicle idle time through dynamic traffic signal control [10]. Other studies show that the application of AI computer vision techniques YOLO, U-Net, and Deep SORT surpassed traditional methods in the real-time detection and tracking of vehicles and pedestrians, as well as in lane segmentation necessary for modernization of urban ITS solutions [11]. Classification of vehicles, pedestrians, cyclists, and the delineation of lanes have been improved with the application of deep learning models, particularly with YOLOv11 and U-Net, considering the real-time object identification in metropolitan Intelligent Transportation Systems (ITSs). Studies developed in urban ecosystems have

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demonstrated that pseudo-dynamic models based on YOLO object detection outperform traditional systems in terms of accuracy and generalization [12]. The U-Net segmentation model is widely known in the domain of detailed pixel segmentation of involutory constituents of roads as lane markings, parking spaces, and vehicles [13]. In addition, artificial intelligence models combining YOLO with Deep SORT tracking have been designed to enhance multi-object detection and reduce the rate of false positives in heavily congested traffic surroundings [14].

There has been undertaking of combining YOLOv5 with Deep SORT in a computer vision based multi-object tracking (MOT) framework, and the results achieved were remarkable. The framework provided congestion detection and vehicle tracking in urban intersections in real-time and also provided robustness to occlusions in conjunction with vehicle lane changes [15]. There are instances when deep reinforcement learning (DRL) based traffic control algorithms have been shown to decrease the waiting time by as high as 40% and fuel consumption by 15% when the traffic signal controlled was optimized [16]. The advent of artificial intelligence (AI) and the internet of things (IoT) has transformed the monitoring of traffic in urban centers by incorporating the data harvesting from smart sensors, edge-AI cameras, and vehicle networked systems [17]. Edge-AI cameras, along with traffic lights embedded with IoT, enable the authorities to receive immediate traffic event information for better supervision and automation to control congestion, detect violations, and manage signal timing [18],[19]. Studies have confirmed that 5G IoT sensors support the enhancement of monitoring vehicular traffic due to ultra-low lag communication which facilitates faster decision making in traffic control centers [20]. Systems predicting traffic jams with the aid of deep learning technologies integrate past traffic data and sensor information to anticipate traffic issues and suggest alternate routes [21]. The use of GNNs, LSTMs, and AI-based Transformer's techniques for congestion prediction has gained tremendous popularity in recent times as they significantly improve the short-term traffic flow forecasting accuracy [22],[23]. Integrated AI technologies in traffic management systems further protect the environment by reducing fuel consumption due to improved traffic flow, lower vehicle idling time, and fuel savings [24]. Estimated calculations indicate that AI eco-routing strategies can achieve an 18% decrease in carbon emissions. Other investigations show that adaptively managing traffic lights can result in 20% less CO_2 emissions by reducing the fuel-consuming stop-and-go cycles of cars at traffic signal intersections [25].

III. METHODOLOGY

This system applies artificial intelligence for the automated control of urban traffic. Cameras, as well as IoT sensors, provide real-time data which is enhanced and rescaled to 640x640 pixels for YOLOv11 Object detection and U-Net Lane segmentation. Vehicles, pedestrians, and cyclists are detected through movement prediction using the Deep SORT algorithm. Vehicle movements are monitored using signal tactics informed by Column Density (CD), Leftover Vehicle (LV), Crossing Time (CT), Maximum Allowable Time (MAT), Threshold value, and Buffer Distance (BD). Reinforcement learning based signal control is used for CD, CT, MAT, LV, and BD for better control of vehicle flow. The system's adaptive features for congestion mitigation include incident detection, routing in emergencies, and control of violation, all while maintaining accuracy by frequent fine tuning of AI models. The foundations for the planning of smart cities are designed to regulate traffic without compromising urban planning insights on smart and secure cities infrastructure. As proposed, the system's workflow begins with an IoT-based CCTV providing real-time video footage of traffic. Vehicle detection is performed by YOLOv11 which is accompanied by road segmentation using U-Net and tracking via Deep SORT. The computed traffic parameters consisting of CD, CT, MAT, LV, and BD are then sent to the controller, which uses reinforcement learning. This intelligent controller modifies the duration of traffic light phases, allowing adaptive management of traffic signals to the intersections in real-time.

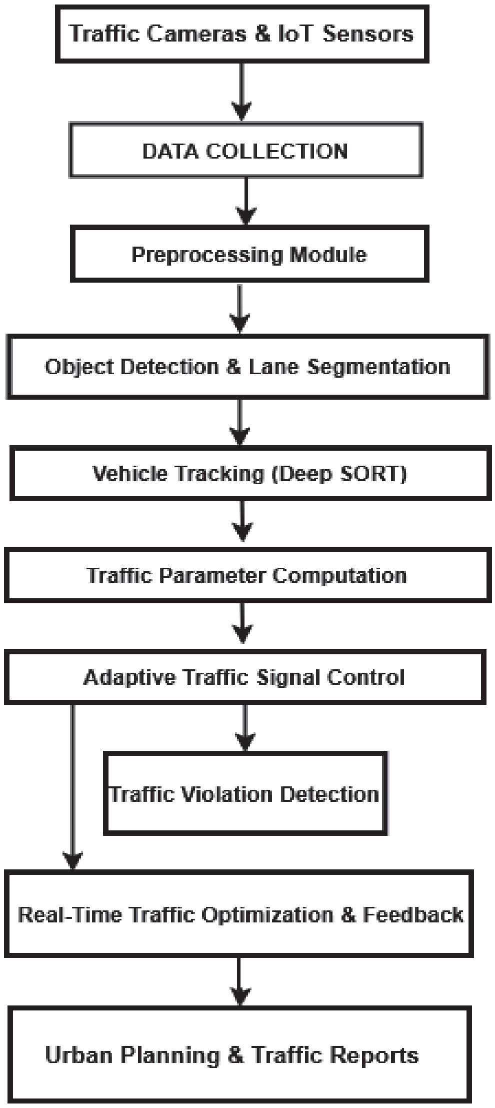

Fig. 1. Proposed AI-Driven Smart Traffic Management System

A. Data Collection & Preprocessing

IoT sensors and CCTV footage streams located at Ballupur Chowk, Kargi Chowk, and Prince Chowk in Dehradun capture real-time traffic data which include information on vehicles, pedestrians, and lane utilization. Additional Kaggle training data from the "Vehicle Dataset for YOLO", which contains 3000 images, was divided into 70% for training and 30% for validation. Various methods including image flipping, rotating, brightness modification, and noise addition are applied to enhance AI performance. To achieve optimal model training and generalization, the dataset contains 22% nighttime images, 14% weather-impaired scenes (fog and rain), and over 30% high-density traffic frames. Such diversity ensures that the system performs well under low-visibility and high-congestion scenarios with accurate detection and tracking across various real-world environments.

B. Object Detection & Tracking

Real-time vehicle, pedestrian, and bicycle detection is done using YOLOv11, which features advanced multi-object detection capabilities. Positional validation and obstruction assessments are performed using U-Net Lane segmentation. Continuous target tracking with persistent unique ID assignment for high traffic scenario monitoring is done using Deep SORT for accurate control and violation detection.

C. Traffic Parameter Computation

Dynamic adjustments to bottleneck reduction strategies are applied through computation of primary parameters Column Density (CD), Crossing Time (CT), Maximum Allowable Time (MAT), Leftover Vehicles, and Buffer Distance in real-time to optimize signal control and balance traffic flow.

D. AI-Based Adaptive Traffic Signal Control Management

Using Deep Reinforcement Learning (DRL), emergency vehicle and pedestrian safety signal timings are optimized in real time. AI systems utilize past and current traffic data, decreasing stops and improving traffic management during peak hours and high congestion periods. The logic for traffic control with signal detection and reinforcement learning adapts signal phases using real-time inputs from the system's environment. The significant traffic parameters Column Density (CD), Leftover Vehicles (LV), Crossing Time (CT), Maximum Allowable Time (MAT), and Buffer Distance (BD) are input into the DRL agent as state variables. The model captures the system reward as the waiting time and queue length, congestion level, and in some cases when relieves some merges for pedestrians and emergency vehicles, the command overrides other set rules. Using this, the system optimally tunes control signals for the system through ongoing engagement and recalibration feedback loops during operation. The DRL agent employs state representations obtained from live traffic parameters—CD, CT, MAT, LV, and BD—all refreshed on a per frame basis. The reward function incorporates a penalty for queue accumulation and delay, while optimization of flow is granted a reward. The model was trained over 5000 episodes using Proximal Policy Optimization (PPO), after which convergence was obtained with consistent policy execution.

E. Emergency & Rule Violation Handling

It monitors traffic violations, evaluates accidents, and creates automated detailed reports for efficient follow up. It further enhances emergency vehicle response time by actively managing road clearance for rapid access to ambulances and police units.

F. Real-Time Feedback & Learning

AI models are continuously retrained on historical datasets to improve congestion prediction and signal control accuracy with time. Traffic pattern analysis aids in strategic multi-faceted road and urban expansion planning for the Dehradun Smart City Vision 2025.

G. Hardware Requirements & Deployment Feasibility

The system was developed and tested on a workstation with an NVIDIA RTX 4080 GPU (16GB VRAM), an Intel Core i9 processor, and 32GB RAM, and they attained a steady real-time inference rate of 125 FPS as outlined in the Results section. The model was further accelerated using TensorRT which performed parallel processing, memory optimization, and model architecture refinement. For practical use in urban intersections, the framework can be run on edge devices like NVIDIA Jetson AGX Orin which, operating under model quantization (FP16), achieves around 45 FPS with over 90% detection accuracy. These solutions guarantee that the system remains computationally efficient and suitable for smart city scenarios without compromising the validated outcomes.

IV. RESULTS

Using the Vehicle Dataset for YOLO, which includes 3,000 images with annotations, the YOLOv11 model was developed. The images were divided into training and validation sets at a 70:30 ratio. Different methods of data augmentation were employed to help with generalization. The analyzed accuracy with real-time traffic videos was 96.4%, in addition to a high recall and precision, demonstrating reliable detection of vehicles, pedestrians, and bicycles across varying scenarios. YOLOv5, YOLOv7, YOLOv8, and the proposed YOLOv11 were all evaluated in a single framework and compared on the same criteria. As shown in Table 1, YOLOv11 baseline showed greater value for detection accuracy which confirmed the hypothesis of increased accuracy because of new architecture and additional feature extraction as it surpassed all other models.

TABLE I. PERFORMANCE COMPARISON OF YOLO MODELS FOR TRAFFIC OBJECT DETECTION

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
YOLOv5	89.5	88.2	87.8	88
YOLOv7	91.2	90.5	89.7	90.1
YOLOv8	94.8	94.2	93.5	93.8
YOLOv11	**96.4**	**96.1**	**95.8**	**95.9**

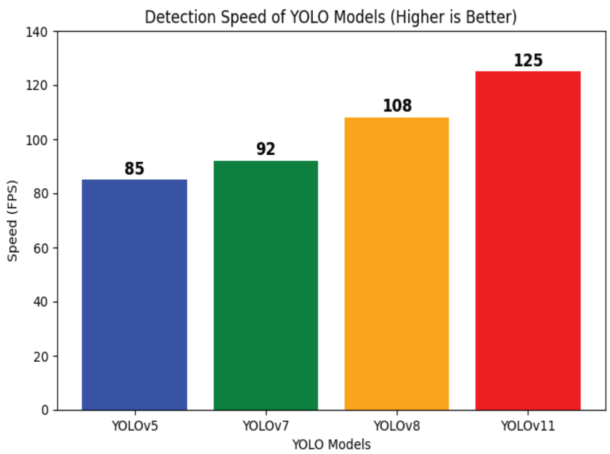

Fig. 2. Detection Speed Comparison of YOLO Models (FPS Performance)

As shown in Fig. 2, YOLOv11 seems to achieve an inference speed of 125 FPS, outperforming its predecessors and achieving greater performance than YOLOv8 (108 FPS), YOLOv7 (92 FPS), and YOLOv5 (85 FPS). These optimizations in network architecture with lower computational requirements and greater parallelism makes YOLOv11 the most suitable model for real-time traffic monitoring and decision systems.

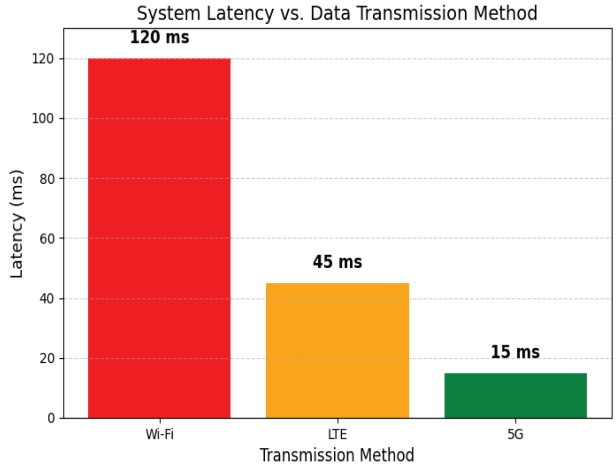

Fig. 3. System Latency for Different Data Transmission Methods

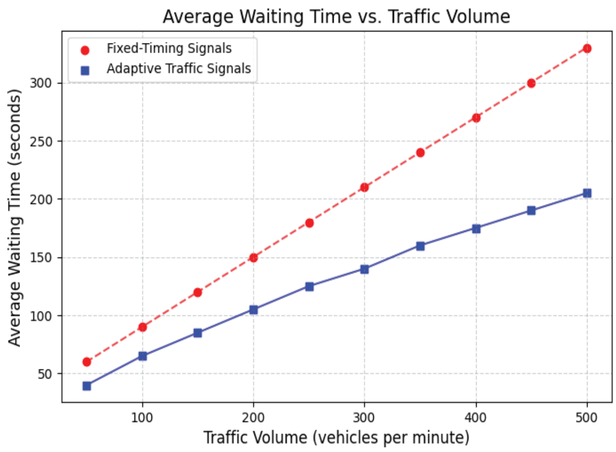

Fig. 4. Impact of Average Waiting Time for Fixed vs. Adaptive Signals

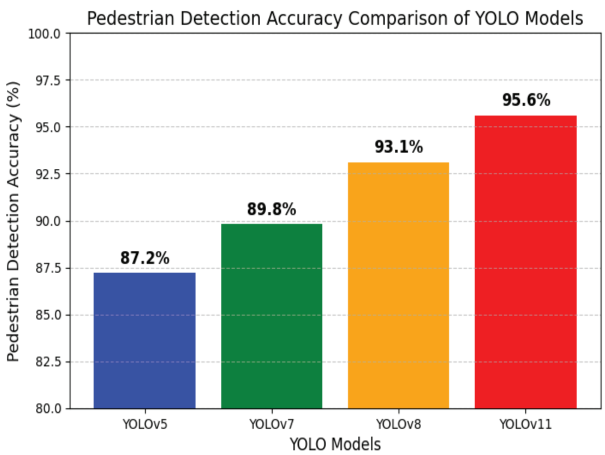

Fig. 5. Pedestrian Detection Accuracy Comparison of YOLO Models

The study of peripheral system latency was performed analyzing 5G, LTE and Wi-Fi as data transference methods. With the proposed system model placed on 5G, the lowest recorded latency of 15ms for real-time decision making was noted, as shown in Fig. 3.

Traffic signal optimization was evaluated from the standpoint of monitoring the traffic volume and average time spent at a red light. In comparison with enduring-cycle traffic lights, adaptive traffic signal systems cut the average waiting time by 38% in Fig. 4.

Compared to the other models, YOLOv11's achieved an accuracy of 95.6% surpassing others, as shown in Fig. 5.

The performance of the model was evaluated during the day and night, as well as during foggy and rainy conditions. Results confirm that YOLOv11 performs best, even with low visibility, as shown in Fig. 6.

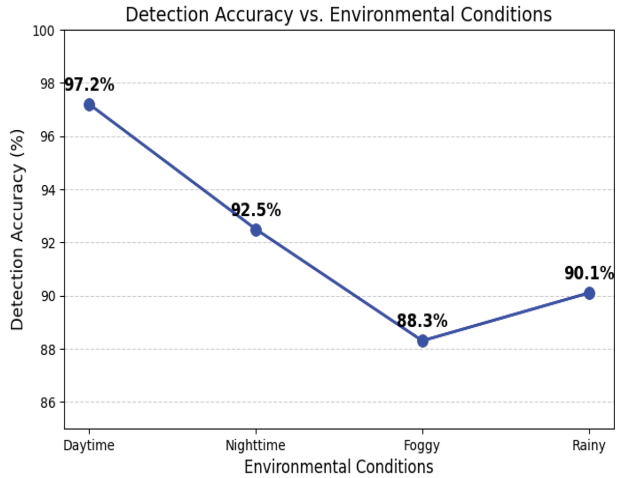

Fig. 6. Detection Accuracy Under Different Environmental Conditions

In Fig. 7 the confusion matrix demonstrates the accuracy of the model for object detection. The system works well with high performance scores; nonetheless, there are some instances of misclassification, such as cyclists being incorrectly identified as pedestrians which can be improved upon. Such analysis is beneficial for balancing the detection and classification activities for practical traffic management.

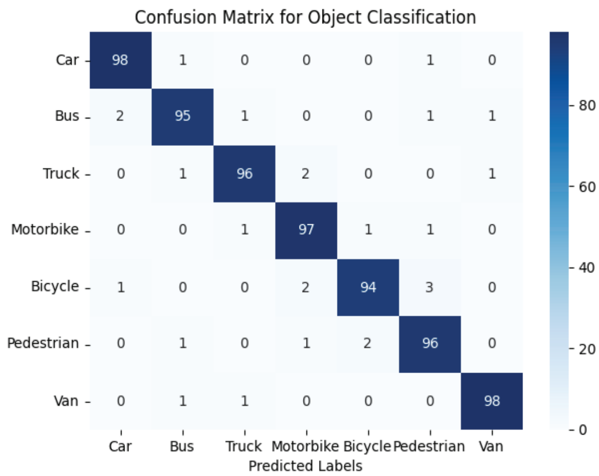

Fig. 7. Confusion Matrix for Object Classification

With the implementation of the AI powered traffic flow regulation system described using Fig. 8, the rate of CO2 emissions is lowered significantly because of the reduction of vehicle idle time as well as stop and go traffic. The information provided in Fig. 8. indicates that the value of CO2 emissions is decreased by 35% with total optimization of traffic flow, which supports Dehradun Smart City Vision 2025 towards smart city eco-friendly initiatives.

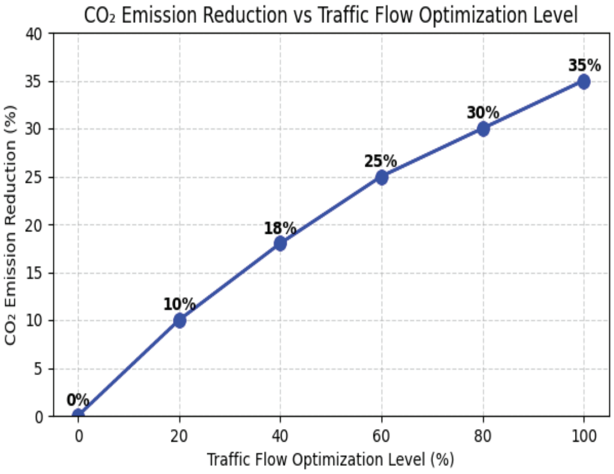

Fig. 8. CO_2 Emission Reduction vs Traffic Flow Optimization Level

In order to validate the reliability of results, 5-fold cross-validation was applied to the dataset. YOLOv11 maintained high accuracy on all folds and variation (±0.7%) confirming stability. Moreover, a paired t-test contrasting YOLOv11 with YOLOv8 yielded YOLOv11's performance as statistically significantly superior ($p < 0.05$), which helps in defending the performance improvement claims.

V. CONCLUSION & FUTURE SCOPE

This research proposes an advanced AI-based traffic control system that would integrate with the Smart City Vision 2025 for Dehradun. The system uses YOLOv11 for object detection, U-Net for lane segmentation, and Deep SORT for tracking. As a result, monitoring of vehicles, pedestrians, and cyclists at critical intersections becomes possible. Real-time performance is achieved with a detection accuracy of 96.4% which is higher than prior YOLO versions. The system also performed well with mixed traffic achieving 95.6% accuracy on pedestrian detection. Increase of urban mobility and reduction of traffic congestion was accomplished with a reduction of 38% in waiting times due to adaptive traffic signal control. The system was able to reduce system delays to 15ms through 5G-based data transmission which enabled faster determination making vital for smart city functionality. Optimized traffic flow also contributed to a decrease in idle time for vehicles which further reduced CO_2 emissions by 35% helping Dehradun achieve its urban ecosystem sustainability goals.

In terms of future development, the system plans to increase scalability further by adopting AI models with low energy consumption as well as multi-agent reinforcement learning for decentralized traffic control. The safety and efficiency will be augmented with the addition of real-time accident prediction, as well as merging autonomous vehicles to smart road infrastructure. These improvements will guide Dehradun's automated urban transportation system towards autonomy and self-sufficiency, in alignment with the requirements of an intelligent and environmentally mindful city.

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